AegisCare HMS — Database Schema & System Architecture

Document Title: Database Schema Specification & System Topology
Document Version: 1.0.0
Database Engine: PostgreSQL 15+ (Supabase Managed with RLS)
Target Environment: Kenya MoH Level 1–6 Facilities

Section 1 — Entity Relationship Diagram (Mermaid ERD)

```
erDiagram
    %% ==========================================
    %% PATIENT CORE & ADMISSIONS
    %% ==========================================
    PATIENTS ||--o{ ENCOUNTERS : "has"
    PATIENTS ||--o{ PATIENT_CONSENTS : "grants"
    PATIENTS ||--o{ CONSENT_OTPS : "receives"
    PATIENTS ||--o{ APPOINTMENTS : "books"
    PATIENTS ||--o{ ADMISSIONS : "undergoes"
    PATIENTS ||--o{ INVOICES : "billed"
    PATIENTS ||--o{ MORTUARY_RECORDS : "admitted_postmortem"

    WARDS ||--o{ BEDS : "contains"
    WARDS ||--o{ ADMISSIONS : "hosts"
    BEDS ||--o{ ADMISSIONS : "occupies"
    ENCOUNTERS ||--o{ ADMISSIONS : "triggers"

    %% ==========================================
    %% CLINICAL & DIAGNOSTICS
    %% ==========================================
    ENCOUNTERS ||--o{ CLINICAL_NOTES : "documents"
    ENCOUNTERS ||--o{ ENCOUNTER_DIAGNOSES : "coded_with"
    ENCOUNTERS ||--o{ LAB_ORDERS : "orders"
    ENCOUNTERS ||--o{ RADIOLOGY_ORDERS : "orders"
    ENCOUNTERS ||--o{ PRESCRIPTIONS : "prescribes"
    ENCOUNTERS ||--o{ ENCOUNTER_ROOM_VISITS : "routes"
    ENCOUNTERS ||--o{ ENCOUNTER_INDICATOR_TAGS : "tagged"

    LAB_ORDERS ||--o{ LAB_RESULTS : "yields"
    LAB_TEST_CATALOG ||--o{ LAB_ORDERS : "defines"
    RADIOLOGY_ORDERS ||--o{ RADIOLOGY_RESULTS : "yields"
    ADMISSIONS ||--o{ MEDICATION_ADMINISTRATIONS : "administers"
    PRESCRIPTIONS ||--o{ MEDICATION_ADMINISTRATIONS : "schedules"

    %% ==========================================
    %% SPECIALIZED UNITS (ICU & DIALYSIS)
    %% ==========================================
    ADMISSIONS ||--o{ ICU_HOURLY_CHARTS : "tracks_hourly"
    ENCOUNTERS ||--o{ DIALYSIS_SESSIONS : "conducts"
    MACHINES ||--o{ DIALYSIS_SESSIONS : "operates_on"
    MACHINES ||--o{ MACHINE_LOGS : "logs_service"

    %% ==========================================
    %% STOCK & PHARMACY
    %% ==========================================
    STOCK_LOCATIONS ||--o{ STOCK_MOVEMENTS : "stores"
    STOCK_LOCATIONS ||--o{ STOCK_TRANSFERS : "transfers_from_or_to"
    STOCK_LOCATIONS ||--o{ STOCK_USAGE : "consumes_from"
    STOCK_ITEMS ||--o{ STOCK_MOVEMENTS : "adjusts"
    STOCK_ITEMS ||--o{ STOCK_TRANSFERS : "transferred"
```

```
    STOCK_ITEMS ||--o{ STOCK_USAGE : "used"
    STOCK_ITEMS ||--o{ PRESCRIPTIONS : "fulfills"

    %% ==========================================
    %% BILLING & INVOICING
    %% ==========================================
    ENCOUNTERS ||--o{ INVOICES : "provisions"
    INVOICES ||--o{ INVOICE_LINE_ITEMS : "contains"
    INVOICES ||--o{ INVOICE_PAYMENTS : "settled_by"
    MORTUARY_RECORDS ||--o{ INVOICES : "billed_via"

    %% ==========================================
    %% INSURANCE, SHA & HIE INTEROPERABILITY
    %% ==========================================
    INSURANCE_PROVIDERS ||--o{ INSURANCE_BENEFIT_PLANS : "offers"
    INSURANCE_PROVIDERS ||--o{ CONTRACTED_PRICES : "negotiates"
    INSURANCE_BENEFIT_PLANS ||--o{ INSURANCE_BENEFIT_CATEGORIES : "defines_benefits"
    ENCOUNTERS ||--o{ SHA_CLAIMS : "generates_claim"
    SHA_CLAIMS ||--o{ SHA_CLAIM_ITEMS : "itemizes"
    ENCOUNTERS ||--o{ DHA_OUTBOUND_QUEUE : "queues_sync"
    ENCOUNTERS ||--o{ EPISODE_OF_CARE : "grouped_by"

    %% ==========================================
    %% AUTH, ROLES & ROOMS
    %% ==========================================
    PROFILES ||--o{ USER_ROLES : "assigned"
    ROOMS ||--o{ ENCOUNTER_ROOM_VISITS : "visited"
    ROOMS ||--o{ USER_ROOM_ACCESS : "grants_access"
    ROOMS ||--o{ STOCK_LOCATIONS : "houses_inventory"

    %% ==========================================
    %% ENTITY DEFINITIONS & ATTRIBUTES
    %% ==========================================
    PATIENTS {
        uuid id PK
        string file_number UK
        string first_name
        string family_name
        string national_id
        date date_of_birth
        boolean dob_known
        string sex
        string phone
        string county
        jsonb next_of_kin
        boolean is_deceased
        timestamp created_at
    }

    ENCOUNTERS {
        uuid id PK
        uuid patient_id FK
        string encounter_type
        string payment_mode
        uuid insurance_provider_id FK
        string insurance_policy_number
        string status
        boolean is_emergency
        jsonb vitals
        uuid current_room_id FK
        timestamp created_at
        timestamp updated_at
    }

    ADMISSIONS {
        uuid id PK
        uuid encounter_id FK
        uuid patient_id FK
        uuid ward_id FK
        uuid bed_id FK
        string status
        timestamp admitted_at
        timestamp discharged_at
        text discharge_summary
        string admitting_doctor
```

```
    }

    WARDS {
        uuid id PK
        string name
        string ward_type
        numeric daily_rate
        integer capacity
        boolean is_active
    }

    BEDS {
        uuid id PK
        uuid ward_id FK
        string bed_number
        string status
    }

    CLINICAL_NOTES {
        uuid id PK
        uuid encounter_id FK
        uuid admission_id FK
        string note_type
        text content
        uuid authored_by FK
        timestamp authored_at
    }

    ENCOUNTER_DIAGNOSES {
        uuid id PK
        uuid encounter_id FK
        string icd11_code
        string icd11_title
        string icd11_uri
        string diagnosis_type
        integer sequence
    }

    LAB_ORDERS {
        uuid id PK
        uuid encounter_id FK
        uuid patient_id FK
        uuid catalog_id FK
        string priority
        string status
        uuid ordered_by FK
        timestamp ordered_at
    }

    LAB_RESULTS {
        uuid id PK
        uuid order_id FK
        jsonb result_data
        string qualitative_result
        string interpretation
        string status
        uuid verified_by FK
        timestamp verified_at
    }

    RADIOLOGY_ORDERS {
        uuid id PK
        uuid encounter_id FK
        uuid patient_id FK
        string modality
        string anatomical_site
        string priority
        string status
    }

    RADIOLOGY_RESULTS {
        uuid id PK
        uuid order_id FK
        text findings
        text impression
```

```
    string radiologist
    jsonb image_paths
    timestamp reported_at
}

PRESCRIPTIONS {
    uuid id PK
    uuid encounter_id FK
    uuid stock_item_id FK
    string drug_name
    string dosage
    string frequency
    string duration
    numeric quantity
    string status
    uuid dispensed_by FK
    timestamp dispensed_at
}

MEDICATION_ADMINISTRATIONS {
    uuid id PK
    uuid admission_id FK
    uuid prescription_id FK
    string drug_name
    string dose_given
    string route
    uuid administered_by FK
    timestamp administered_at
}

ICU_HOURLY_CHARTS {
    uuid id PK
    uuid admission_id FK
    integer hour_number
    numeric arterial_bp_systolic
    numeric arterial_bp_diastolic
    numeric cvp
    integer gcs_total
    integer rass_score
    numeric peep
    numeric fio2_percent
    numeric urine_output_ml
    jsonb inotropes
    uuid logged_by FK
    timestamp recorded_at
}

DIALYSIS_SESSIONS {
    uuid id PK
    uuid encounter_id FK
    uuid patient_id FK
    uuid machine_id FK
    numeric pre_weight_kg
    numeric post_weight_kg
    numeric uf_goal_ml
    numeric uf_achieved_ml
    numeric blood_flow_rate
    numeric dialysate_flow_rate
    string dialyzer_model
    string access_type
    jsonb complications
    string status
    timestamp session_start
    timestamp session_end
}

MORTUARY_RECORDS {
    uuid id PK
    string mortuary_reference UK
    string body_source
    string deceased_name
    string gender
    date date_of_death
    string storage_slot
    string brought_in_by_name
```

```
        string police_ob_number
        uuid released_to_national_id
        timestamp admitted_at
        timestamp released_at
    }

    STOCK_ITEMS {
        uuid id PK
        string name
        string category
        string unit
        numeric current_quantity
        numeric reorder_level
        numeric unit_price
        numeric cash_price
        numeric insurance_price
    }

    STOCK_LOCATIONS {
        uuid id PK
        string name
        string code UK
        uuid room_id FK
        boolean is_active
    }
```

```
    STOCK_MOVEMENTS {
        uuid id PK
        uuid item_id FK
        numeric change
        string reason
        string notes
        uuid created_by FK
        timestamp created_at
    }

    STOCK_TRANSFERS {
        uuid id PK
        uuid item_id FK
        uuid from_location_id FK
        uuid to_location_id FK
        numeric quantity
        string status
        uuid initiated_by FK
    }

    STOCK_USAGE {
        uuid id PK
        uuid encounter_id FK
        uuid item_id FK
```

```
        uuid location_id FK
        numeric used_quantity
        string usage_reason
        timestamp used_at
    }

    INVOICES {
        uuid id PK
        string invoice_number UK
        uuid encounter_id FK
        uuid patient_id FK
        numeric subtotal
        numeric discount
        numeric waiver_amount
        numeric amount_paid
        numeric balance
        string status
        timestamp created_at
    }

    INVOICE_LINE_ITEMS {
        uuid id PK
        uuid invoice_id FK
        uuid encounter_id FK
        string item_type
```

```
        uuid source_id
        string description
        numeric quantity
        numeric unit_price
        numeric amount
    }

    INVOICE_PAYMENTS {
        uuid id PK
        uuid invoice_id FK
        numeric amount
        string method
        string reference
        uuid received_by FK
        timestamp paid_at
    }

    INSURANCE_PROVIDERS {
        uuid id PK
        string name
        string code UK
        numeric coverage_percentage
        boolean is_active
    }
```

```
    INSURANCE_BENEFIT_PLANS {
        uuid id PK
        uuid provider_id FK
        string plan_name
        string plan_code
        numeric copay_amount
        integer visit_limit
    }

    SHA_CLAIMS {
        uuid id PK
        uuid encounter_id FK
        uuid patient_id FK
        string claim_type
        string fund_type
        string preauth_code
        numeric claim_amount
        string status
        jsonb fhir_bundle
        timestamp submitted_at
    }

    SHA_CLAIM_ITEMS {
        uuid id PK
        uuid claim_id FK
```

```
        string tariff_code
        string item_description
        numeric unit_price
        numeric claimed_amount
    }

    DHA_OUTBOUND_QUEUE {
        uuid id PK
        uuid encounter_id FK
        uuid patient_id FK
        string queue_type
        string insurer_type
        jsonb payload
        string status
        integer attempts
        timestamp created_at
    }

    PROFILES {
        uuid id PK
        string full_name
        string council_type
        string council_registration_number
        string phone
        string national_id
```

```
        boolean is_approved
    }

    USER_ROLES {
        uuid id PK
        uuid user_id FK
        string role
    }

    ROLE_PERMISSIONS {
        string role PK
        string permission PK
    }

    ROOMS {
        uuid id PK
        string name
        string code
        string kind
        boolean is_active
    }

    MACHINES {
        uuid id PK
        string name
        string model
        string serial_number
        string status
        string location
    }

    MACHINE_LOGS {
        uuid id PK
        uuid machine_id FK
        string log_type
        date log_date
        string description
        numeric cost
    }

    APP_SETTINGS {
        string id PK
        string facility_name
        string facility_kmhfl_code
        string facility_sha_id
        integer facility_level
        string county
    }

    AUDIT_LOG {
        uuid id PK
        string table_name
        uuid record_id
        string action
        jsonb old_data
        jsonb new_data
        uuid changed_by FK
        timestamp changed_at
    }
```

Section 2 — Patient Journey Flows (Mermaid Flowcharts)

Flow 1: Cash Outpatient Journey

```
flowchart TD
    A[Patient Arrives at Facility] --> B[Reception: Patient Lookup / Registration]
    B --> C[Consent Acquisition: OTP SMS via Africa's Talking]
    C --> D[Select Payment Mode: Cash]
    D --> E[Route Encounter to Triage Room]
    E --> F[Triage: Vital Signs, BMI, Acuity Scoring]
    F --> G[Route to Doctor Consultation Room]
    G --> H[Doctor Consultation: Clinical Notes & ICD-11 Coding]

    H --> I{Diagnostic Workup Required?}
    I -- Yes: Lab Tests --> J[Lab Order Generated & Billed]
    J --> K[Cashier: Pay Lab Invoice Items]
    K --> L[Lab: Specimen Collection & Result Entry]
    L --> M[Lab Tech Verifies: SMS Sent to Patient]
    M --> H

    I -- Yes: Radiology --> N[Radiology Order Generated & Billed]
    N --> O[Cashier: Pay Imaging Invoice Items]
    O --> P[Radiology: Scan & Radiologist Report]
    P --> Q[Report Signed: SMS Sent to Patient]
    Q --> H

    I -- No --> R[Doctor Issues Prescription]
    R --> S[Doctor Electronically Signs Encounter - Lock Records]
    S --> T[Encounter Tagged for MOH 705A/B Demographics]
    T --> U[Cashier: Settle Final Invoice Balance]
    U --> V[Receipt Issued & SMS Payment Confirmation]
    V --> W[Pharmacy: Dispense Drugs from Pharmacy Store]
    W --> X[Patient Discharged / Departure]
```

Flow 2: Insurance Inpatient Journey

```
flowchart TD
    A[Patient Arrives with Private / Corporate Insurance] --> B[Reception: Register & Link Underwriter Policy]
    B --> C[Consent Verification: OTP SMS]
    C --> D[Route to Insurance Desk]
    D --> E[Insurance Agent: Verify Eligibility & Pre-Authorization]
    E --> F[Contracted Tariffs Applied to Encounter]
    F --> G[Triage: Baseline Vitals Capture]
    G --> H[Doctor Consultation: Examination & Diagnosis]
    H --> I[Doctor Orders Inpatient Ward Admission]

    I --> J[Assign Inpatient Ward & Specific Bed]
    J --> K[Bed Status Set to Occupied]
    K --> L[Ward Nursing Care: Medication Administrations MAR]
    L --> M[Nightly pg_cron Daemon: Accrue Daily Bed Charges]

    M --> N{Patient Clinically Ready for Discharge?}
    N -- No --> L
    N -- Yes --> O[Doctor Completes Mandatory Discharge Summary]
    O --> P[Doctor Signs Admission File]
    P --> Q[Pharmacy: Reconcile & Dispense Discharge Medications]
    Q --> R[Billing: Compile Itemized Interim Inpatient Bill]
    R --> S[Insurance Desk: Submit Electronic Claim / Guarantee]
    S --> T[Underwriter Approval Received / Copay Paid]
    T --> U[Discharge Clearance Granted & Bed Set to Available]
    U --> V[Patient Discharged Home]
```

Flow 3: SHA SHIF Journey (Including Biometric Step)

```
flowchart TD
    A[Patient Presents for SHA Care] --> B[Reception: Search National ID / SHA Number]
    B --> C[Consent Capture: OTP SMS Verification]
    C --> D[Select Payment Mode: sha_shif / phf / eccif]
    D --> E[Insurance Desk: Query SHA Member Registry API ■]
    E --> F[Biometric Smart-Card / Fingerprint Verification ■]
    F --> G[SHA Fund Type Classified: PHF / SHIF / ECCIF]
    G --> H[Triage: Vitals Scoring & MoH Demographic Tagging]
    H --> I[Doctor Consultation: WHO ICD-11 Diagnosis Search]

    I --> J[Doctor Prescribes Drugs & Diagnostic Workups]
    J --> K[SHA Standard Benefit Packages & Tariffs Linked]
    K --> L[Doctor Electronically Signs Encounter]

    L --> M[Database Trigger: auto_generate_sha_claim]
    M --> N[RPC: build_fhir_claim Compiles FHIR Claim Resource]
    N --> O[Edge Function: claims-dispatcher Routes Payload]
    O --> P[Queued in dha_outbound_queue with mediator_id]
    P --> Q[DHA AfyaLink HIE / SHA Claims Gateway ■]
    Q --> R[Pharmacy Dispenses SHIF-Covered Medications]
    R --> S[Patient Discharged]
```

Flow 4: ICU Admission Journey

```
flowchart TD
    A[Critically Ill Patient in Casualty / Operating Theatre] --> B[Doctor Orders Emergency ICU Admission]
    B --> C[Assign Bed in Intensive Care Unit Ward]
    C --> D[RPC: charge_icu_admission_fee Auto-Posts Admission Tariff]
    D --> E[Bed Occupancy Set to Occupied in ICU Ward]
    E --> F[ICU Nursing Station: Continuous Patient Monitoring]

    F --> G[Hourly Flow-Sheet Entry: Arterial BP, CVP, ABG]
    G --> H[Ventilator Parameters: Mode, PEEP, FiO2, VT]
    H --> I[Neurological: GCS /15 & RASS Sedation Scoring]
    I --> J[Bedside Drug Dispensing: Deduct from ICU Store]

    J --> K[Nightly pg_cron: Accrue Daily ICU Charges]
    K --> L{Patient Hemodynamically Stable?}
    L -- No --> F
    L -- Yes: Transfer to General Ward --> M[Trigger: audit_ward_transfers Logs Transfer]
    M --> N[Bed Status in ICU Reset to Cleaning/Available]
    N --> O[Patient Continues Care in General Ward]
```

Flow 5: Dialysis Session Journey

```
flowchart TD
    A[Renal Patient Arrives for Scheduled Hemodialysis] --> B[Reception / Dialysis Unit Check-In]
    B --> C[Pre-Dialysis Assessment: Pre-Weight, Dry Weight, Access Site]
    C --> D[Dialysis Nurse Assigns Dialysis Machine Number]
    D --> E[Dialyzer Prescription: UF Goal mL, BFR, DFR, Treatment Time]
    E --> F[Select Session Consumables: Bloodlines, Needles, Acid/Bicarb]

    F --> G[Initiate Hemodialysis Run]
    G --> H[Hourly Monitoring: Intra-dialytic BP, Pulse, TMP, Net UF]
    H --> I[Session Concluded: Post-Weight & Complications Logged]

    I --> J[RPC: process_dialysis_session_billing Executed]
    J --> K[Deduct Consumables from Dialysis Store in stock_usage]
    K --> L[Post Itemized Dialysis Session Charge to Invoice]
    L --> M{Payment Channel}
    M -- SHA SHIF Renal Package --> N[Queue Renal Claim in dha_outbound_queue]
    M -- Private Insurance --> O[Insurance Clearance & Voucher Settlement]
    M -- Cash --> P[Cashier: Pay Dialysis Invoice]
    N --> Q[Patient Discharged with Follow-Up Schedule]
    O --> Q
    P --> Q
```

Flow 6: Mortuary (External Body) Journey

```
flowchart TD
    A[Deceased Brought in Dead: Police Case / Home Death] --> B[Mortuary Department Intake]
    B --> C[Mortician Captures Intake Metadata & Police OB Number]
    C --> D[Trigger: assign_mortuary_reference Generates Reference Number]
    D --> E[Assign Cold Storage Refrigeration Slot]
    E --> F[RPC: create_external_mortuary_invoice Provisions Master Bill]

    F --> G[Preservation / Embalming / Autopsy Pathology Logged]
    G --> H[Nightly pg_cron: accrue_daily_mortuary_charges Posts Daily Fees]

    H --> I[Next-of-Kin Arrives with Burial Permit & National ID]
    I --> J[Cashier: Reconcile & Settle Mortuary Invoice Balance]
    J --> K{Invoice Paid in Full or Authorized Waiver?}
    K -- No --> L[System Release Gate: Body Release Locked]
    L --> J
    K -- Yes --> M[Mortician Documents Handover & Vehicle Registration]
    M --> N[Mortuary Record Status Set to Released]
    N --> O[Cold Storage Slot Reset to Available]
```

Section 3 — System Architecture Diagram (Mermaid)

```
flowchart TB
    subgraph CLIENT_LAYER ["Frontend Client Tier (Browser)"]
        UI["React 18 Single-Page Application"]
        ROUTER["TanStack Router (Type-Safe Routing)"]
        QUERY["TanStack Query (Optimistic State & Cache)"]
        SHADCN["Tailwind CSS + Shadcn UI Component Suite"]
        UI --- ROUTER --- QUERY --- SHADCN
    end

    subgraph HOSTING_EDGE ["Hosting & CDN Tier"]
        VERCEL["Vercel Edge Network / Lovable Cloud"]
        BUN["Bun Runtime & Build Tooling"]
        VERCEL --- BUN
    end

    subgraph SUPABASE_BACKEND ["Supabase Cloud Platform"]
        AUTH["Supabase Auth (JWT, Role Mapping)"]
        REALTIME["Supabase Realtime (WebSocket Queues & Beds)"]

        subgraph POSTGRES_DB ["PostgreSQL 15 Database Engine"]
            TABLES["35+ Relational Relational Tables"]
            RLS["Row-Level Security (RLS) Policies"]
            TRIGGERS["PL/pgSQL Business Triggers & Rules"]
            CRON["pg_cron Background Scheduling Daemons"]
            AUDIT["Append-Only Audit Logging & Archive"]
```

```
            TABLES --- RLS --- TRIGGERS --- CRON --- AUDIT
        end

        subgraph EDGE_FUNCTIONS ["Supabase Deno Edge Functions"]
            SMS_FN["send-sms (Africa's Talking Gateway)"]
            ICD_FN["icd11-search (WHO Cloud Proxy)"]
            DISPATCH_FN["claims-dispatcher (HIE & Claims Router)"]
            FHIR_PAT["fhir-patient (R4 Resource Builder)"]
            FHIR_ENC["fhir-encounter (R4 Resource Builder)"]
            FHIR_COND["fhir-condition (R4 Resource Builder)"]
            FHIR_BND["fhir-bundle (R4 Collection Builder)"]
        end
    end

    subgraph EXTERNAL_INTEGRATIONS ["External Gateways & Regulators"]
        AT_SMS["Africa's Talking SMS API (Production ■)"]
        WHO_API["WHO ICD-11 MMS Cloud API (Production ■)"]
        DHA_HIE["Kenya DHA AfyaLink HIE (Pending Credentials ■)"]
        SHA_GATEWAY["SHA SHIF / PHF / ECCIF Claims (Pending ■)"]
        MPESA["Safaricom M-Pesa Daraja Gateway (Planned ■)"]
    end

    UI -->|HTTPS / REST & GraphQL| POSTGRES_DB
    UI -->|WSS Realtime Feeds| REALTIME
    UI -->|JWT Auth Handshake| AUTH
```

```
    UI -->|Edge Invocations| EDGE_FUNCTIONS

    SMS_FN -->|HTTPS REST| AT_SMS
    ICD_FN -->|OAuth2 / REST| WHO_API
    DISPATCH_FN -->|FHIR R4 REST| DHA_HIE
    DISPATCH_FN -->|FHIR Claim REST| SHA_GATEWAY
    POSTGRES_DB -.->|Nightly Cron Execution| CRON
```

Section 4 — Table Descriptions & Security Schema

1. `patients`

- **Purpose:** Master Patient Index (MPI) repository storing demographic identity, contact details, next-of-kin, and vital status.
- **Key Columns:** `id` (UUID PK), `file_number` (Unique Text, e.g., `P000142`), `first_name`, `middle_name`, `family_name`, `patient_name`, `national_id`, `date_of_birth`, `dob_known` (Boolean), `estimated_age`, `sex` (`male`/`female`), `phone`, `email`, `county`, `next_of_kin` (JSONB), `is_deceased` (Boolean).
- **Relationships:** One-to-many with `encounters`, `admissions`, `invoices`, `appointments`, `patient_consents`, `consent_otps`.
- **RLS Policy Summary:**
- `SELECT`: Permitted for all authenticated and approved hospital staff (`is_approved(auth.uid())`).
- `INSERT` / `UPDATE`: Restricted to users possessing `register_patient`, `records_create`, or `admin` permissions.
- `DELETE`: Explicitly prohibited (`USING (false)`).

2. `encounters` (Aliased as `patient_registrations`)

- **Purpose:** Central operational table tracking patient episodes spanning OPD registration, triage vitals, consultation, and discharge.
- **Key Columns:** `id` (UUID PK), `patient_id` (FK `patients.id`), `encounter_type` (`outpatient`, `inpatient`, `emergency`), `payment_mode` (`cash`, `insurance`, `sha_shif`, `exemption`), `insurance_provider_id` (FK `insurance_providers.id`), `insurance_policy_number`, `status` (`waiting`, `in_progress`, `done`, `signed`, `cancelled`), `is_emergency` (Boolean), `vitals` (JSONB), `current_room_id` (FK `rooms.id`), `subtotal`, `amount_paid`, `created_at`, `updated_at`.
- **Relationships:** Belongs to `patients`; Has many `clinical_notes`, `encounter_diagnoses`, `lab_orders`, `radiology_orders`, `prescriptions`, `invoices`, `admissions`, `sha_claims`.
- **RLS Policy Summary:**
- `SELECT`: Permitted for approved staff.
- `INSERT`: Permitted for reception, triage, and clinical staff.

- `UPDATE`: Permitted for attending clinicians and room staff; mutations strictly blocked once `status = 'signed'` via `enforce_encounter_lock` trigger.

---

3. `admissions`

- **Purpose:** Tracks inpatient hospitalizations from bed allocation through clinical ward care to discharge summary.
- **Key Columns:** `id` (UUID PK), `encounter_id` (FK `encounters.id`), `patient_id` (FK `patients.id`), `ward_id` (FK `wards.id`), `bed_id` (FK `beds.id`), `admitted_at`, `discharged_at`, `admitting_doctor`, `admission_reason`, `admission_type`, `status` (`admitted`, `discharged`, `transferred`), `discharge_summary` (Text).
- **Relationships:** Belongs to `encounters`, `patients`, `wards`, `beds`; Has many `icu_hourly_charts`, `medication_administrations`.
- **RLS Policy Summary:**
- `SELECT`: Approved staff.
- `INSERT` / `UPDATE`: Restricted to clinical staff (`doctor`, `clinical_officer`, `nurse`, `admin`). Discharge requires non-null `discharge_summary`.

---

4. `wards` & `beds`

- **Purpose:** Physical inpatient infrastructure management, tracking bed capacities, occupancy states, and daily ward rates.
- **Key Columns:**
- `wards`: `id` (UUID PK), `name`, `ward_type` (`general`, `maternity`, `pediatric`, `surgical`, `icu`, `hdu`), `daily_rate` (Numeric), `capacity` (Integer), `is_active` (Boolean).
- `beds`: `id` (UUID PK), `ward_id` (FK `wards.id`), `bed_number` (Text), `status` (`available`, `occupied`, `maintenance`, `cleaning`).
- **Relationships:** `wards` has many `beds` and `admissions`. `beds` belongs to `wards`.
- **RLS Policy Summary:**
- `SELECT`: Approved staff.
- `INSERT` / `UPDATE` / `DELETE`: Restricted to `admin` role.

---

5. `clinical_notes`

- **Purpose:** Structured clinical documentation entered by attending doctors and clinical officers.
- **Key Columns:** `id` (UUID PK), `encounter_id` (FK `encounters.id`), `admission_id` (FK `admissions.id`), `note_type` (`chief_complaint`, `hpi`, `ros`, `examination`, `plan`, `progress_note`), `content` (Text), `authored_by` (FK `auth.users`), `authored_at`.
- **Relationships:** Belongs to `encounters`, `admissions`.
- **RLS Policy Summary:**
- `SELECT`: Approved medical and nursing staff.
- `INSERT`: Attending clinicians. Updates locked once parent encounter is signed.

---

6. `encounter_diagnoses`

- **Purpose:** International diagnostic disease codes linked to encounters, mapped to WHO ICD-11 MMS terminology.
- **Key Columns:** `id` (UUID PK), `encounter_id` (FK `encounters.id`), `icd11_code` (Varchar), `icd11_title` (Text), `icd11_uri` (Text), `diagnosis_type` (`primary`, `secondary`, `working`, `differential`), `sequence` (Integer), `notes` (Text).
- **Relationships:** Belongs to `encounters`.
- **RLS Policy Summary:**
- `SELECT`: Approved clinical staff.
- `INSERT` / `UPDATE`: Attending clinicians. Protected by `clean_and_validate_diagnosis_insert()` and locked upon encounter signing.

---

7. `lab_orders` & `lab_results`

- **Purpose:** Laboratory diagnostic requisitions, analyzer result records, reference range validation, and technologist verification.
- **Key Columns:**
- `lab_orders`: `id` (UUID PK), `encounter_id`, `patient_id`, `catalog_id` (FK `lab_test_catalog.id`), `priority` (`routine`, `urgent`), `status` (`ordered`, `in_progress`, `completed`, `cancelled`), `ordered_by`.
- `lab_results`: `id` (UUID PK), `order_id` (FK `lab_orders.id`), `result_data` (JSONB), `qualitative_result` (Text), `interpretation` (Text), `status` (`draft`, `verified`), `verified_by` (FK `auth.users`), `verified_at`.
- **Relationships:** `lab_orders` belongs to `encounters`; has one `lab_results`.
- **RLS Policy Summary:**
- `lab_orders`: Clinicians insert; Lab techs update status.
- `lab_results`: `lab_tech` role has exclusive insert/update rights. Verified results are immutable.

---

8. `radiology_orders` & `radiology_results`

- **Purpose:** Medical imaging workflow management from clinician requisition to radiologist diagnostic report.
- **Key Columns:**
- `radiology_orders`: `id` (UUID PK), `encounter_id`, `patient_id`, `modality` (`xray`, `ultrasound`, `ct`, `mri`, `ecg`), `anatomical_site`, `priority`, `status`, `clinical_indication`.
- `radiology_results`: `id` (UUID PK), `order_id` (FK `radiology_orders.id`), `findings` (Text), `impression` (Text), `radiologist` (Text), `image_paths` (JSONB), `reported_at`.
- **Relationships:** `radiology_orders` belongs to `encounters`; has one `radiology_results`.
- **RLS Policy Summary:**
- `radiology_orders`: Clinicians insert; Radiologists update.
- `radiology_results`: Restricted to `radiologist` and `doctor` roles.

---

9. `prescriptions` & `medication_administrations`

- **Purpose:** Clinical pharmaceutical orders, pharmacy dispensing validations, and inpatient bedside nursing administration (MAR).
- **Key Columns:**
- `prescriptions`: `id` (UUID PK), `encounter_id` (or `registration_id`), `stock_item_id` (FK `stock_items.id`), `drug_name`, `dosage`, `frequency`, `duration`, `quantity`, `status` (`pending`, `dispensed`, `cancelled`), `dispensed_by`, `dispensed_at`.
- `medication_administrations`: `id` (UUID PK), `admission_id`, `prescription_id`, `drug_name`, `dose_given`, `route`, `administered_by`, `administered_at`.
- **Relationships:** `prescriptions` belongs to `encounters`, `stock_items`; has many `medication_administrations`.
- **RLS Policy Summary:**
- `prescriptions`: Clinicians create; `pharmacist` updates to `dispensed` (triggering stock decrement).
- `medication_administrations`: Restricted to `nurse` and `doctor` roles.

---

10. `icu_hourly_charts`

- **Purpose:** High-frequency critical care monitoring records for ICU inpatients.
- **Key Columns:** `id` (UUID PK), `admission_id` (FK `admissions.id`), `hour_number` (Integer), `arterial_bp_systolic`, `arterial_bp_diastolic`, `cvp`, `gcs_total` (3–15), `rass_score` (-5 to +4), `peep`, `fio2_percent`, `urine_output_ml`, `inotropes` (JSONB), `logged_by`, `recorded_at`.
- **Relationships:** Belongs to `admissions`.
- **RLS Policy Summary:**
- `SELECT`: Approved ICU clinical staff.
- `INSERT` / `UPDATE`: Restricted to `nurse`, `doctor`, and `admin` roles in ICU room.

---

11. `dialysis_sessions`

- **Purpose:** Clinical flow-sheet for hemodialysis treatments and automated consumable billing.
- **Key Columns:** `id` (UUID PK), `encounter_id`, `patient_id`, `machine_id` (FK `machines.id`), `pre_weight_kg`, `post_weight_kg`, `uf_goal_ml`, `uf_achieved_ml`, `blood_flow_rate`, `dialysate_flow_rate`, `dialyzer_model`,

access_type, complications (JSONB), status, session_start, session_end.
- **Relationships:** Belongs to encounters, patients, machines.
- **RLS Policy Summary:**
- Restricted to nurse, doctor, and admin assigned to the Dialysis Unit.

---

12. mortuary_records

- **Purpose:** Administrative tracking of deceased bodies admitted internally or externally.
- **Key Columns:** id (UUID PK), mortuary_reference (Unique Text, e.g., MORT-2026-0012), body_source (internal, external), deceased_name, gender, date_of_death, storage_slot, brought_in_by_name, police_ob_number, released_to_national_id, admitted_at, released_at.
- **Relationships:** Has one linked invoices record.
- **RLS Policy Summary:**
- Restricted to mortician, accountant, and admin roles. Body release blocked if invoice has an unpaid balance.

---

13. stock_items, stock_locations, stock_movements, stock_transfers, stock_usage

- **Purpose:** Real-time multi-location perpetual inventory management across Main Warehouse, Pharmacy, ICU, and Dialysis stores.
- **Key Columns:**
- stock_items: id (UUID PK), name, category, unit, current_quantity, reorder_level, unit_price, cash_price, insurance_price.
- stock_locations: id (UUID PK), name, code (Unique), room_id (FK rooms.id), is_active.
- stock_movements: id (UUID PK), item_id, change (Numeric), reason (delivery, dispense, transfer_in, transfer_out, usage, adjustment), notes, created_by, created_at.
- stock_transfers: id (UUID PK), item_id, from_location_id, to_location_id, quantity, status.
- stock_usage: id (UUID PK), encounter_id, item_id, location_id, used_quantity, usage_reason.
- **Relationships:** stock_items has many movements, transfers, and usage records.
- **RLS Policy Summary:**
- Read: Approved staff.
- Mutations: Governed by user_stock_location_access matrix and executed via secure RPCs (apply_stock_movement trigger).

---

14. invoices, invoice_line_items, invoice_payments

- **Purpose:** Master fiscal ledger maintaining real-time itemized billing, payments, waivers, and aging.
- **Key Columns:**
- invoices: id (UUID PK), invoice_number (Unique Text, e.g., INV-2026-00891), encounter_id, patient_id, subtotal, discount, waiver_amount, amount_paid, balance, status (draft, pending, partial, paid, waived, claimed).
- invoice_line_items: id (UUID PK), invoice_id, encounter_id, item_type (consultation, lab, radiology, prescription, bed_day, procedure, consumable), source_id, description, quantity, unit_price, amount.
- invoice_payments: id (UUID PK), invoice_id, amount, method (cash, mpesa, card, insurance, bank), reference, received_by, paid_at.
- **Relationships:** invoices has many invoice_line_items and invoice_payments.
- **RLS Policy Summary:**
- Select: Approved staff.
- Payments / Waivers: Restricted to accountant and admin roles. Line items automatically maintained by database triggers.

---

15. sha_claims & sha_claim_items

- **Purpose:** Electronic claim repository for Social Health Authority (SHA) reimbursement packages.
- **Key Columns:**
- sha_claims: id (UUID PK), encounter_id, patient_id, claim_type (outpatient, inpatient), fund_type (phf, shif, eccif), preauth_code, claim_amount, status (draft, pending, submitted, approved, rejected), fhir_bundle (JSONB),

submitted_at.
- sha_claim_items: id (UUID PK), claim_id (FK sha_claims.id), tariff_code, item_description, unit_price, claimed_amount.
- **Relationships:** Belongs to encounters, patients; has many sha_claim_items.
- **RLS Policy Summary:**
- Read: Approved staff.
- Insert / Update: Restricted to insurance_agent, accountant, and admin roles.

---

16. dha_outbound_queue

- **Purpose:** Asynchronous outbound message queue for DHA AfyaLink HIE and SHA FHIR bundles.
- **Key Columns:** id (UUID PK), encounter_id, patient_id, queue_type (fhir_sync, sha_claim, private_claim, cash_receipt), insurer_type, payload (JSONB), status (pending, processing, completed, failed, skipped), attempts (Integer), error_message, created_at.
- **Relationships:** Belongs to encounters, patients.
- **RLS Policy Summary:**
- Select: Approved administrative staff.
- Mutations: Managed exclusively by Edge Functions via service_role key.

---

17. profiles, user_roles, role_permissions, user_room_access

- **Purpose:** RBAC security framework enforcing access control and practitioner credential verification.
- **Key Columns:**
- profiles: id (UUID PK, references auth.users.id), full_name, council_type (KMPDC, COC, NCK, PPB, KMLTTB), council_registration_number, national_id, phone, is_approved (Boolean).
- user_roles: id (UUID PK), user_id (FK auth.users), role (app_role enum).
- role_permissions: role (app_role), permission (Text PKs).
- user_room_access: user_id (FK auth.users), room_id (FK rooms.id).
- **RLS Policy Summary:**
- profiles: Users can read/update their own profile; admins can approve accounts.
- user_roles & role_permissions: Exclusively manageable by admin role.

---

18. app_settings & facility_features

- **Purpose:** Global institutional settings and dynamic facility-level feature switches.
- **Key Columns:** id (Text PK 'global'), facility_name, facility_kmhfl_code, facility_sha_id, facility_level (1–6), county, address, phone, email.
- **RLS Policy Summary:**
- Select: Public/authenticated.
- Update: Exclusively restricted to admin and system_admin roles.

---

19. audit_log, audit_log_archive, audit_archive_runs

- **Purpose:** Tamper-evident operational audit trail and long-term 20-year compliance archive.
- **Key Columns:**
- audit_log: id (UUID PK), table_name, record_id, action (INSERT, UPDATE, DELETE, BREAK_GLASS), old_data (JSONB), new_data (JSONB), changed_by (FK auth.users), changed_at.
- audit_log_archive: Mirror schema of audit_log with archived_at timestamp.
- audit_archive_runs: id, run_at, rows_archived, status, error_message.
- **RLS Policy Summary:**
- Append-Only Security: SELECT permitted for approved admins; INSERT, UPDATE, DELETE strictly blocked for all user JWTs (USING (false)). Mutations occur strictly via SECURITY DEFINER database triggers.